
Impact on Small Business Loans During the 2008 Recession

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Abstract

We aim to look at small business loans sent out between 2007-2012, and see if we can determine if ZIP code increases the likelihood of loans being approved and repaid. This is an important problem for us as this was the last major economic recession in US history, and as graduate students we are curious to know what may be a good way to ride out another potential recession.

1 Introduction

Many people have a fear of a recession as of late, with J.P. Morgan Chase estimating a 40 percent chance of the U.S. economy entering a recession by the end of 2025 (<https://money.usnews.com/investing/articles/recession-2025-what-to-watch-how-to-prepare>). As graduate students, we thought it would be a productive experience to leverage our machine learning knowledge and apply it to a dataset of small business loans, specifically looking at the period of 2007-2012 during the 2008 Recession. This will hopefully give us an edge on applying for jobs, by potentially identifying zip codes that were resilient to the effects of a recession, better helping with our job searches in the future.

To do: add high level explanation of ML approach that we choose to use.

2 Related Works

Deep Learning for mortgage risk explores client behavior patterns by modeling payment dynamics across regions and at the ZIP code level. The study primarily focused on tracking prepayment, foreclosure, and delinquency states.

Using loan-level data from 2005-2011, *Small Business Lending and the Great Recession* analyses the lending patterns to small businesses across regions during the financial crisis.

3 Methodology

We will be preprocessing the data set located at <https://www.kaggle.com/datasets/mirbektoktogaraev/should-this-loan-be-approved-or-denied/data>, and we will be running experiments on an ipynb file. Currently, we are in the process of cleaning the data and performing feature engineering. Once the data is clean and we have finished exploratory data analysis, we will split the data, and scale and run it for different ML models using the training and validation sets. We will then compare and contrast the different data sets and choose one to run on the test set.

4 Experiments

5 Results

6 Conclusion

7 Acknowledgements

For the TAs: is this section meant to contain our bibliography/ works cited?

Cole, Rebel A., and Tatyana Sokolyk. "How Do Banks Respond to Increased Funding Uncertainty? Evidence from the U.S. Mortgage and Small Business Lending Crisis." 2012.

Sirignano, Justin, et al. "Deep Learning for Mortgage Risk." arXiv, 8 July 2016, arxiv.org/abs/1607.02470.